

# Assignment 1 — Simple Linear Regression: TV Advertising vs. Sales

**Author:** Aditya Raj | **Dataset:** tv\_sales\_data.csv (TV budget in \$1,000s, Sales in 1,000s of units, n = 6)

**Model:**  $\text{Sales} = \beta_0 + \beta_1 \cdot \text{TV} + \varepsilon$

**Method:** Coefficients fit via Gradient Descent; standard errors, R-squared, and p-values computed via the exact simple-linear-regression inference formulas applied to the fit's own residuals.

## 1. Code and Method

The model is fit by **Gradient Descent** instead of a closed-form formula: it starts both coefficients at 0, then repeatedly nudges them in the direction that reduces the Mean Squared Error (MSE), using a learning rate of  $\alpha = 0.1$  for 1000 iterations. The TV values are standardized before the loop purely for numerically stable steps, and the learned coefficients are converted back to the original (\$1,000s) TV scale afterward.

```
x = df["TV"].values.astype(float)
y_vals = df["Sales"].values.astype(float)
n = len(x)

x_mean, x_std = x.mean(), x.std()
x_scaled = (x - x_mean) / x_std

b0, b1 = 0.0, 0.0
alpha = 0.1
n_iters = 1000

for i in range(n_iters):
    y_pred = b0 + b1 * x_scaled
    error = y_pred - y_vals
    grad_b0 = (2 / n) * np.sum(error)
    grad_b1 = (2 / n) * np.sum(error * x_scaled)
    b0 -= alpha * grad_b0
    b1 -= alpha * grad_b1

beta1 = b1 / x_std
beta0 = b0 - beta1 * x_mean
```

Gradient Descent only minimizes MSE — it does not, by itself, produce standard errors, R-squared, or p-values. Those are calculated **directly from the fitted line's own residuals**, using the exact simple-linear-regression inference formulas:

```
from scipy import stats

y_fit = beta0 + beta1 * x
residuals = y_vals - y_fit

RSS = np.sum(residuals ** 2)
TSS = np.sum((y_vals - y_vals.mean()) ** 2)
df_resid = n - 2
sigma2 = RSS / df_resid
Sxx = np.sum((x - x_mean) ** 2)

se_beta0 = np.sqrt(sigma2 * (1 / n + x_mean ** 2 / Sxx))
se_beta1 = np.sqrt(sigma2 / Sxx)

t_beta0 = beta0 / se_beta0
t_beta1 = beta1 / se_beta1
p_beta0 = 2 * (1 - stats.t.cdf(abs(t_beta0), df_resid))
```

```
p_beta1 = 2 * (1 - stats.t.cdf(abs(t_beta1), df_resid))
```

```
r_squared = 1 - RSS / TSS
```

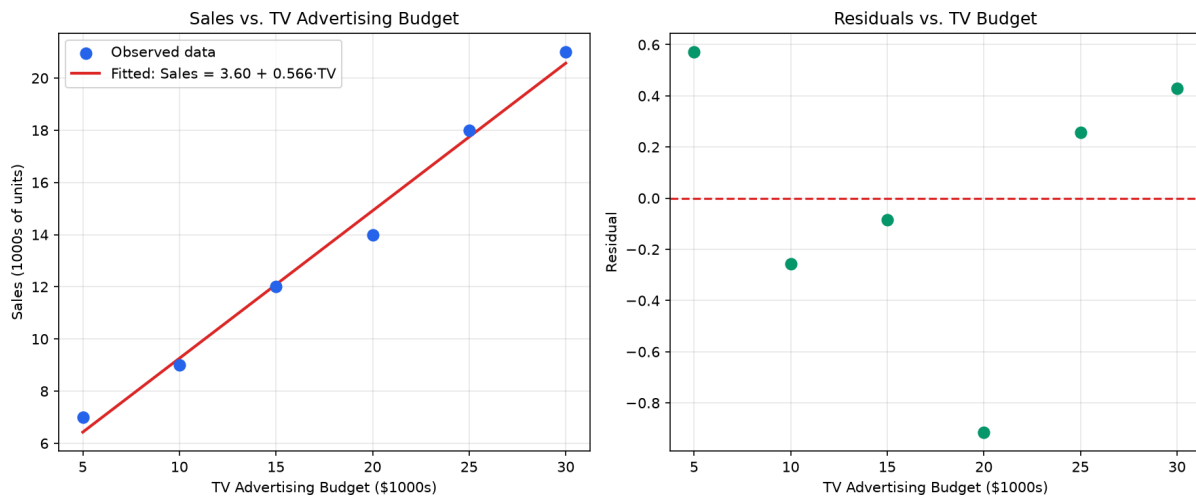
| Quantity          | Formula                                                         |
|-------------------|-----------------------------------------------------------------|
| Residual variance | $\sigma^2 = \text{RSS} / (n - 2)$                               |
| SE(beta1)         | $\sqrt{\sigma^2 / S_{xx}}$ , $S_{xx} = \sum (x_i - \bar{x})^2$  |
| SE(beta0)         | $\sqrt{\sigma^2 * (1/n + \bar{x}^2 / S_{xx})}$                  |
| t-statistic       | $\hat{\beta}_1 / \text{SE}(\hat{\beta}_1)$ , t-dist with n-2 df |
| R-squared         | $1 - \text{RSS} / \text{TSS}$                                   |

## 2. Results

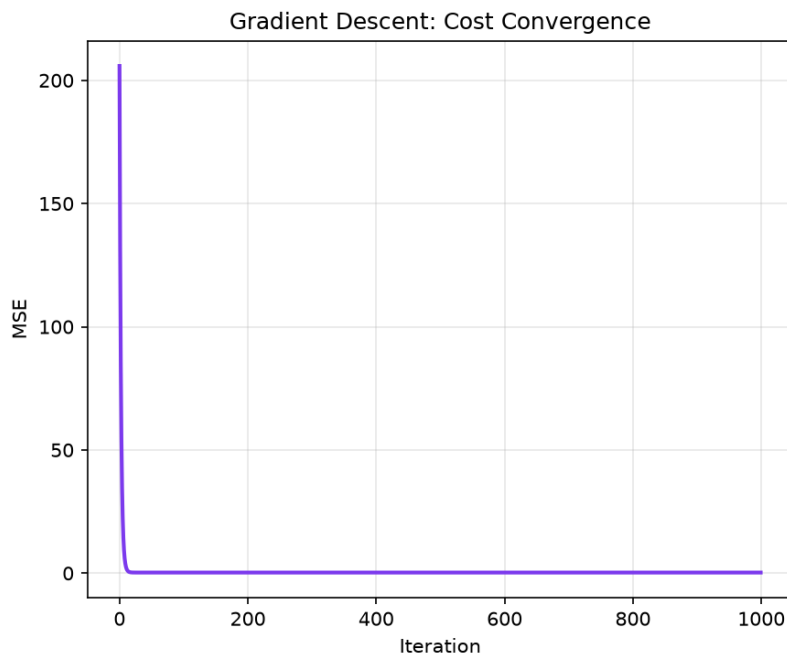
| Quantity            | Value                   |
|---------------------|-------------------------|
| beta0 (intercept)   | 3.6000                  |
| beta1 (slope, TV)   | 0.5657                  |
| SE(beta0)           | 0.5674                  |
| SE(beta1)           | 0.0291                  |
| t-statistic (beta0) | 6.345                   |
| t-statistic (beta1) | 19.415                  |
| p-value (intercept) | 0.003160                |
| p-value (slope)     | $4.15 \times 10^{-5}$   |
| R-squared           | 0.9895                  |
| Degrees of freedom  | 4 (n = 6, 2 parameters) |

Fitted equation: **Sales = 3.60 + 0.5657 x TV**

Fit and residuals:



Gradient Descent cost convergence:



The MSE falls smoothly toward its minimum over 1,000 iterations, confirming the model converged to a stable fit rather than stopping early or diverging.

### 3. Interpretation — Plain Language

- **What beta1 means in everyday terms:**  $\text{beta1} = 0.5657$  is a positive number, so spending more on TV ads goes with more sales. Since TV is measured in \$1,000s and Sales in 1,000s of units, this means: for every extra \$1,000 put into TV advertising, sales go up by roughly 566 units, on average.
- **Is this a real effect, or could it be luck?** The p-value for the slope is 0.0000415 — a tiny number, way below the usual 0.05 cutoff people use to decide if something is "statistically significant." In plain words: it's extremely unlikely we'd see a relationship this strong just by random chance if TV spend actually had no effect on sales. So yes, there's strong evidence TV advertising really does move sales.

- **How much of sales is "explained" by TV budget?** R-squared = 0.9895, meaning about 99% of the ups and downs in sales across these 6 data points can be accounted for just by knowing the TV budget. Only about 1% is left over as unexplained noise. That's an extremely tight fit — expected here because this is a small, clean practice dataset (only 6 points), not noisy real-world data.
- **Bottom line:** the data says TV advertising budget is a strong, reliable, positive driver of sales in this dataset, and a simple straight-line model captures almost all of the pattern.

#### 4. Interpretation — Professional / Technical Language

- **Sign and magnitude of beta1:** The estimated slope,  $\hat{\beta}_1 = 0.5657$ , is positive and indicates a positive linear association between TV advertising expenditure and sales volume. Holding the linear specification fixed, each additional \$1,000 unit of TV budget is associated with an expected increase of approximately 0.5657 thousand units ( $\approx 566$  units) in sales.
- **Statistical significance:** Under  $H_0: \beta_1 = 0$  versus  $H_1: \beta_1 \neq 0$ , the test statistic  $t = \hat{\beta}_1 / \text{SE}(\hat{\beta}_1) = 19.415$  on  $n-2 = 4$  degrees of freedom yields  $p \approx 4.15 \times 10^{-5}$ , far below conventional significance thresholds ( $\alpha = 0.05, 0.01$ ).  $H_0$  is rejected, providing strong statistical evidence of a non-zero linear relationship between TV advertising and sales. The intercept is likewise statistically significant ( $\hat{\beta}_0 = 3.60$ ,  $\text{SE} = 0.5674$ ,  $t = 6.345$ ,  $p \approx 0.0032$ ), though its interpretation as "expected sales at zero TV spend" is a mathematical extrapolation rather than a directly observed operating condition, since the sample range of TV spend is \$5,000-\$30,000.
- **Proportion of variance explained:** R-squared = 0.9895 indicates that approximately 98.95% of the total variability in sales (TSS) is explained by the fitted linear relationship with TV advertising budget, leaving only  $\approx 1.05\%$  attributable to residual (unexplained) variation. Such a high R-squared is atypical for real marketing data and reflects the small sample size ( $n = 6$ ) and the near-linear, low-noise construction of this illustrative dataset rather than a general claim about TV advertising's explanatory power in practice.
- **Overall conclusion:** Within this sample, TV advertising budget is a statistically significant and practically strong positive predictor of sales, and the simple linear model accounts for nearly all of the observed variance in the response.